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USER MANUAL  
SAFECHAT ANTI-GROOMING DETECTION CHAT APPLICATION

THEME: INTELLIGENT ANTI-HARASSMENT SOLUTIONS

FOCUS AREA: IDENTIFYING GROOMING BEHAVIOUR

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# 1.0 Introduction and Setup

The purpose of this manual is to support the setup and use of SafeChat: Anti-Grooming Detection Chat Application for Online Safety, which is designed to provide a secure messaging environment.

Several considerations should be noted when reviewing this document. First, this manual should be read alongside the other materials included in the documentation pack. This report provides recommendations on the types of interactions for which SafeChat is considered appropriate at the time of publication. These recommendations should be carefully reviewed before the application is deployed within any target group or organisation.

Second, SafeChat was tested in simulated chat environments and through a controlled prototype build within the development framework. The accompanying technical documentation and Research Report provide further detail on the testing environment and evaluation scope. These conditions should be considered when assessing the platform's suitability for specific safety programmes.

Finally, the information within this documentation pack may become outdated due to the rapid pace of technological change and advancement. SafeChat is a prototype application that is continuously being updated and improved by its developers. As a result, some screenshots and access methods for specific parts of the application may change over time.

The first section of this guide focuses on navigating the chat interface and monitoring conversations using SafeChat. It includes practical guidance and an overview of the Risk Dashboard, which is used to track potential grooming patterns and flag high-risk interactions in real time.

The second section focuses on the technical setup process, including how SafeChat can be installed on a mobile device by downloading and installing the prototype application package, APK, directly from GitHub.

The final section introduces the newly integrated Safety Hub feature, which includes an AI Awareness Chatbot and built-in reporting tools. This feature is designed to support users and supervisors in managing incident reports and accessing safety resources.

## 1.1 System Requirements

SafeChat is currently deployed as a functional prototype rather than a production-ready application available on commercial app stores; your phone must meet the requirements stated below to run the application.

### 1.1.1 Minimum & Recommended Specifications

Minimum Requirements:

- **Operating System:** Android 14.0
- **Memory Allocation (RAM):** 2 GB of RAM
- **Internal Storage Space:** 6 GB of free storage
- **Display Properties:** Standard modern screen
- **Network Connection:** Stable Wi-Fi connection / 4G Cellular Data .
- **Network Latency:** <150ms Latency
- **Required Device Permissions:** Explicit authorization to **"Install Unknown Apps"** from your browser or file manager, storage read/write permissions and active internet permissions.

Recommended Requirements:

- **Operating System:** Android 14.0 or newer
- **Memory Allocation (RAM):** 4 GB of RAM or higher
- **Internal Storage Space:** 10+ GB of free storage
- **Display Properties:** HD Resolution screen or higher
- **Network Connection:** High-speed Wi-Fi / 5G Cellular Data
- **Network Latency:** 50ms Latency for smooth and quick Gemini analysis
- **Required Device Permissions:** Explicit authorization to **"Install Unknown Apps"** from your browser or file manager, storage read/write permissions and active internet permissions.

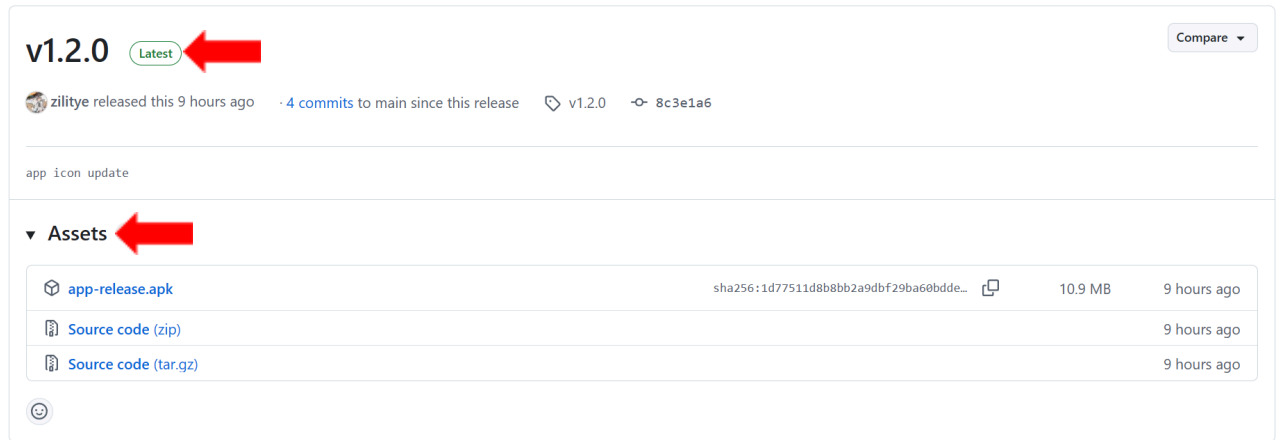
## 1.2 SafeChat Installation

Since SafeChat is currently in its prototype phase, there is no official commercial app store presence. Instead, you'll download the app file (APK) from our GitHub page. Below is a written step-by-step guide for setting up SafeChat on an Android device with accompanying screenshots.

### Step 1: Downloading the Prototype APK

1. Open your mobile browser or testing environment browser and go to the link given below:  
<https://github.com/zilitye/Anti-Grooming-Detection-ChatApp/releases/latest/>
2. Make sure the word 'latest' is stated next to the version number.

[Releases](#) / v1.2.0



v1.2.0 **Latest** Compare

zilitye released this 9 hours ago · 4 commits to main since this release · v1.2.0 · 8c3e1a6

app icon update

▼ **Assets**

|                                                                                                                        |                                        |                                                                                     |         |             |
|------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------|---------|-------------|
|  <a href="#">app-release.apk</a>      | sha256:1d77511d8b8bb2a9dbf29ba60bde... |  | 10.9 MB | 9 hours ago |
|  <a href="#">Source code (zip)</a>    |                                        |                                                                                     |         | 9 hours ago |
|  <a href="#">Source code (tar.gz)</a> |                                        |                                                                                     |         | 9 hours ago |

3. Scroll down and tap 'Assets' to see the file list.
4. Find the file called **app-release.apk**.



|                                                                                                                     |                                        |                                                                                       |         |             |
|---------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------|---------|-------------|
|  <a href="#">app-release.apk</a> | sha256:1d77511d8b8bb2a9dbf29ba60bde... |  | 10.9 MB | 9 hours ago |
|---------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------|---------|-------------|

5. Click directly on **app-release.apk** to download it on your local device.

## Step 2: Grant Permissions to Install Unknown Apps

1. Open your phone's **Settings** app.



2. Navigate to **Privacy Protection > Special Permissions > Install Unknown Apps**

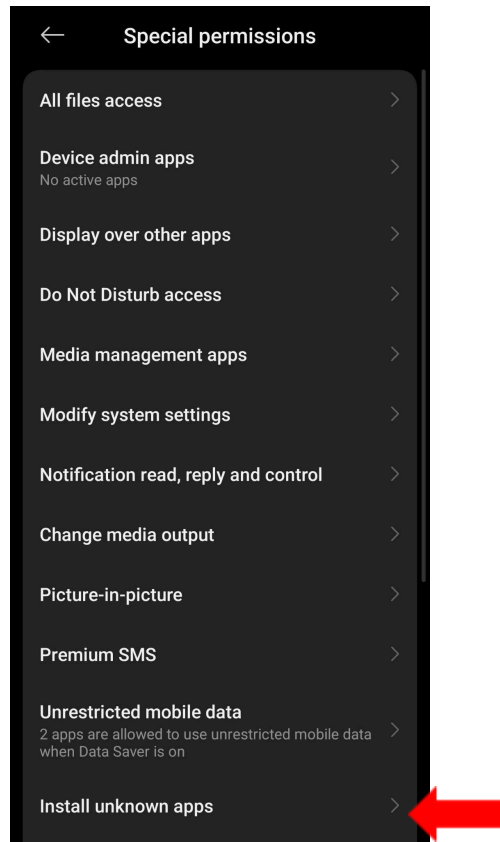


Figure 1 Privacy Settings

3. Select your browser or the **File Manager** app from the list.

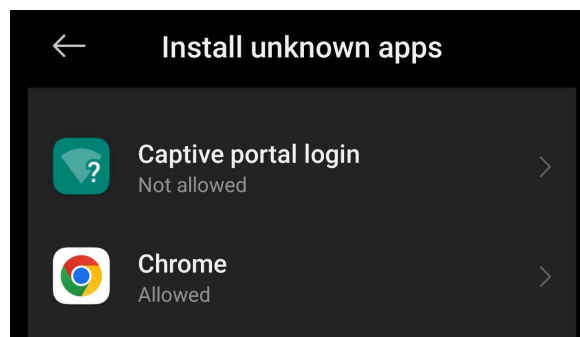


Figure 2 File Manager to Enable App Download

4. Toggle the switch to **Allow from this source**.

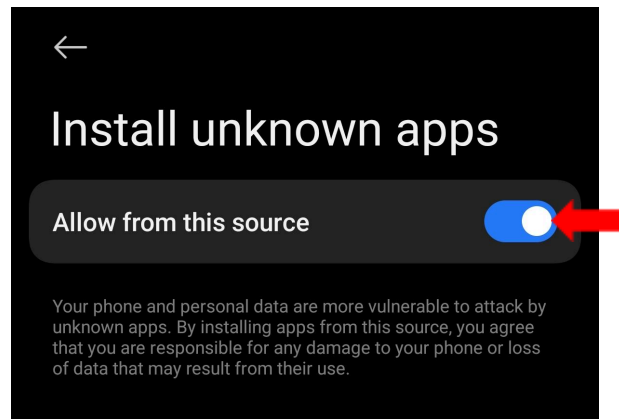


Figure 3 Toggle Bar Allowing Unknown App Installation

Note: Xiaomi/MIUI devices may show an additional 10-second countdown warning. The user shall check the box that says "I am *aware of the possible risks...*" and tap **OK**.

### Step 3: Package Installation

1. Open your File Manager app, or your browser's Downloads list
2. Click on the **app-release.apk** file you just downloaded
3. Select **Install** when the install dialog shows up. Once the installation process successfully completes, tap **Open** to launch SafeChat.

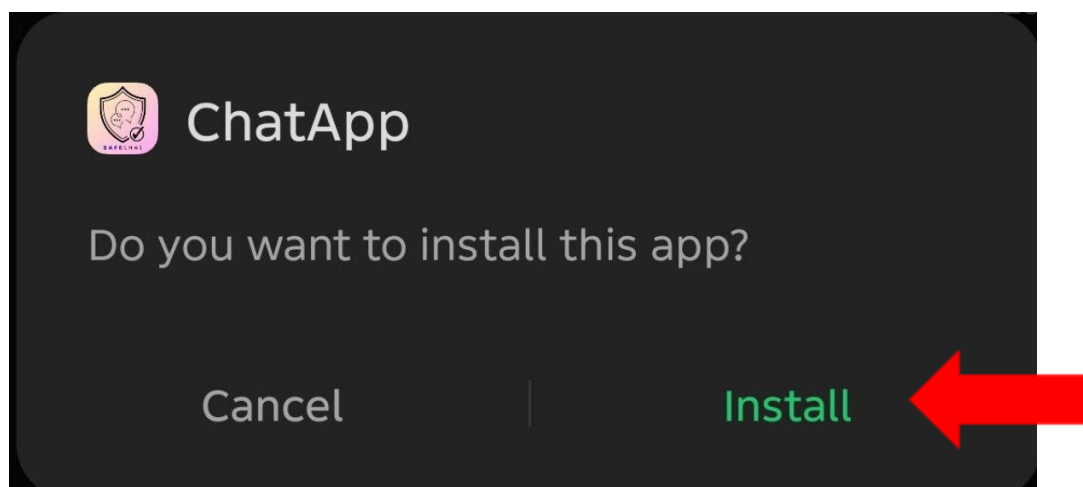


Figure 4 Confirmation Message for App Installation

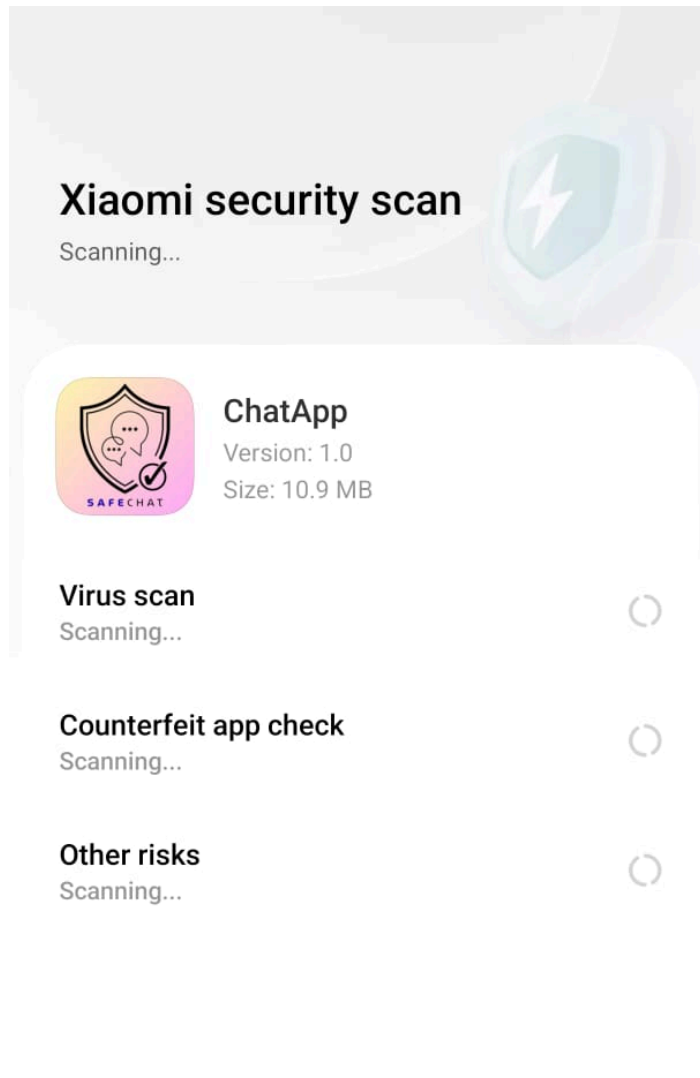


Figure 5 App Scanning for Safety Risk



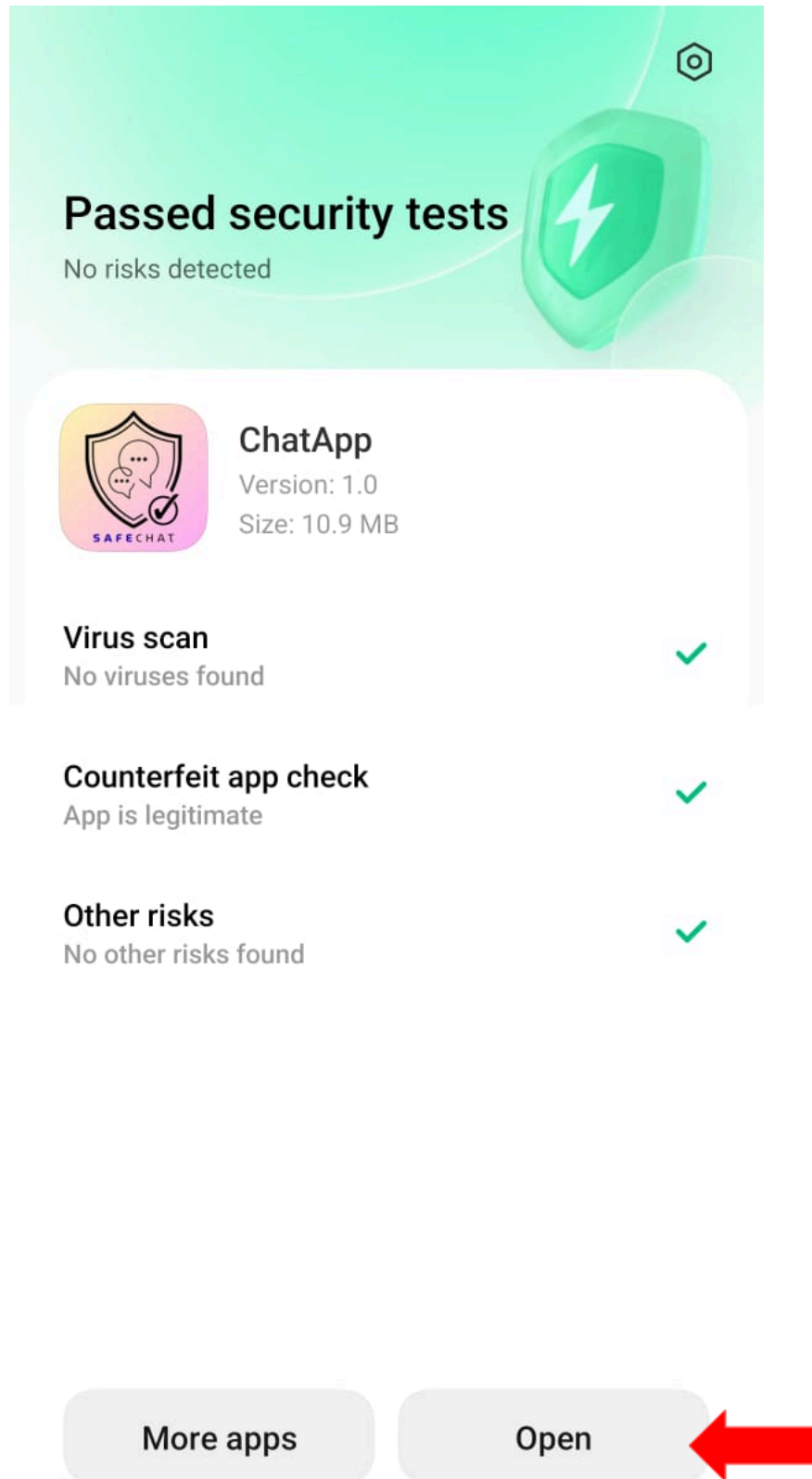


Figure 6 Passed Security Check

## 1.3 Basic Navigation

Once authenticated, navigating **SafeChat** is straightforward and relies on three primary interface layouts.

### 1.3.1 Active Conversation Hub

Upon logging in, you are greeted by your main dashboard. The top header displays your active user profile name (e.g., Chong Yu Ming). This screen shows all your ongoing chats. Each chat shows the contact's photo, name, and last message. To initiate a brand new chat, tap the blue floating **Plus (+) Button** located in the bottom right corner.

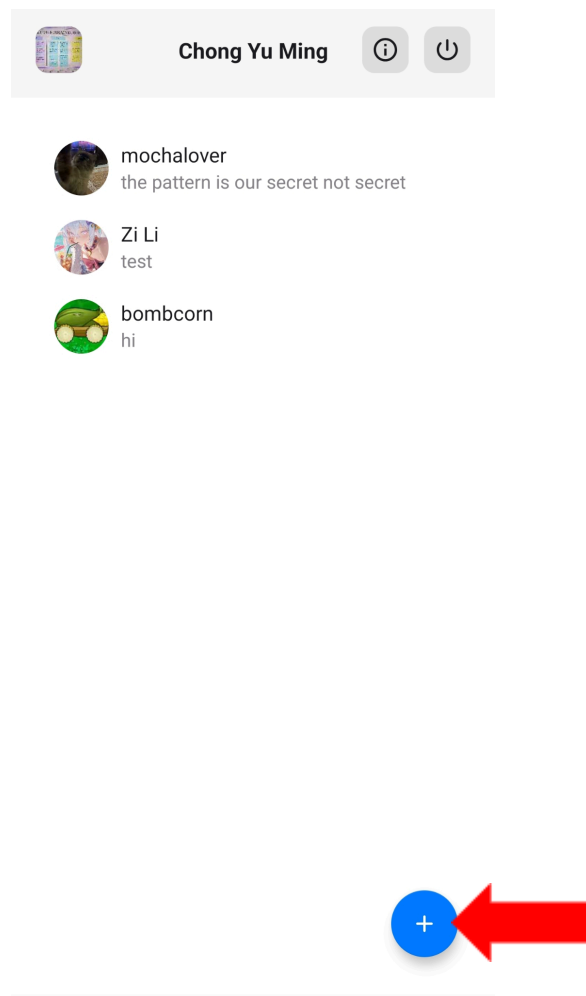


Figure 7 User Main Dashboard

### 1.3.2 Select User Directory

Tapping the **Plus (+) Button** redirects you to the **Select User** screen. This shows a list of everyone you can chat with their avatars, names, and emails. Simply tap any user profile to instantly open a chat window with them.

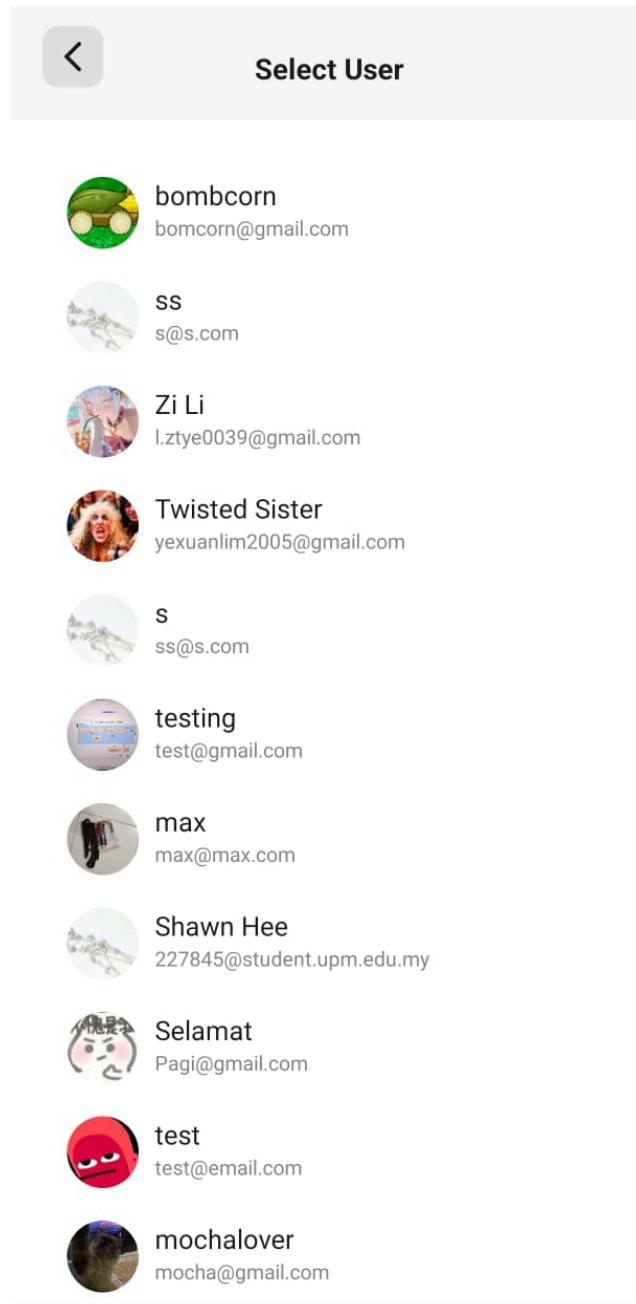


Figure 8 Select User to Chat

### 1.3.3 Inside the Chat Interface

The chat room window is split into three functional zones:

- **The Top Header:** Contains a **Back Arrow (<)** to return to the recent chat, the contact's name, and an **Information (i) Icon** which opens the Risk Dashboard
- **The Chat Stream:** Displays incoming messages in grey bubbles on the left and your sent messages in blue bubbles on the right. Messages flagged for potential grooming risks will display a red **Warning Triangle (Δ)** next to the bubble.
- **The Text Input Bar:** Located at the very bottom, featuring a Type a message field and a blue circular send arrow.

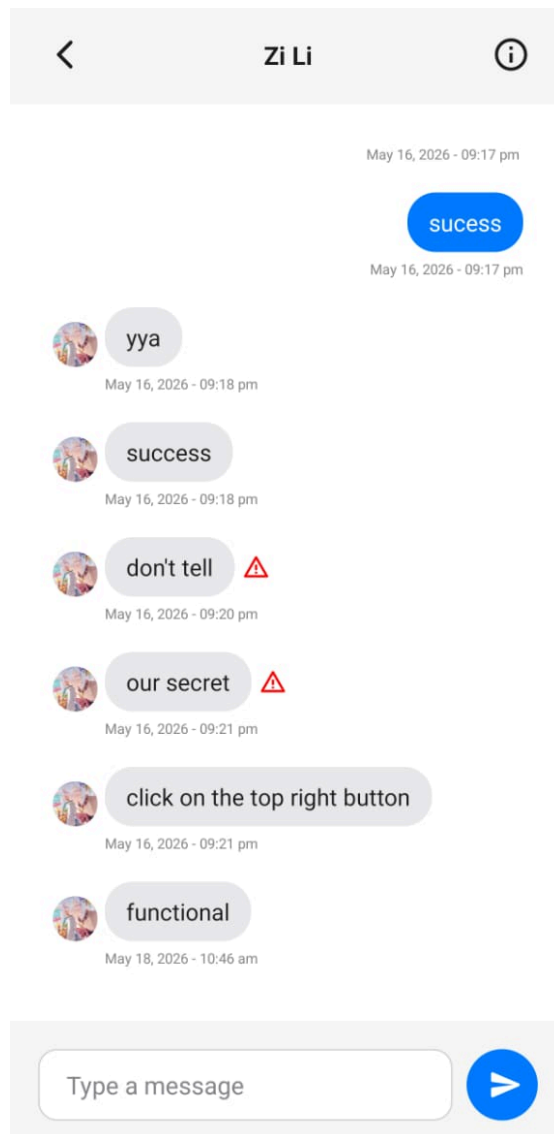
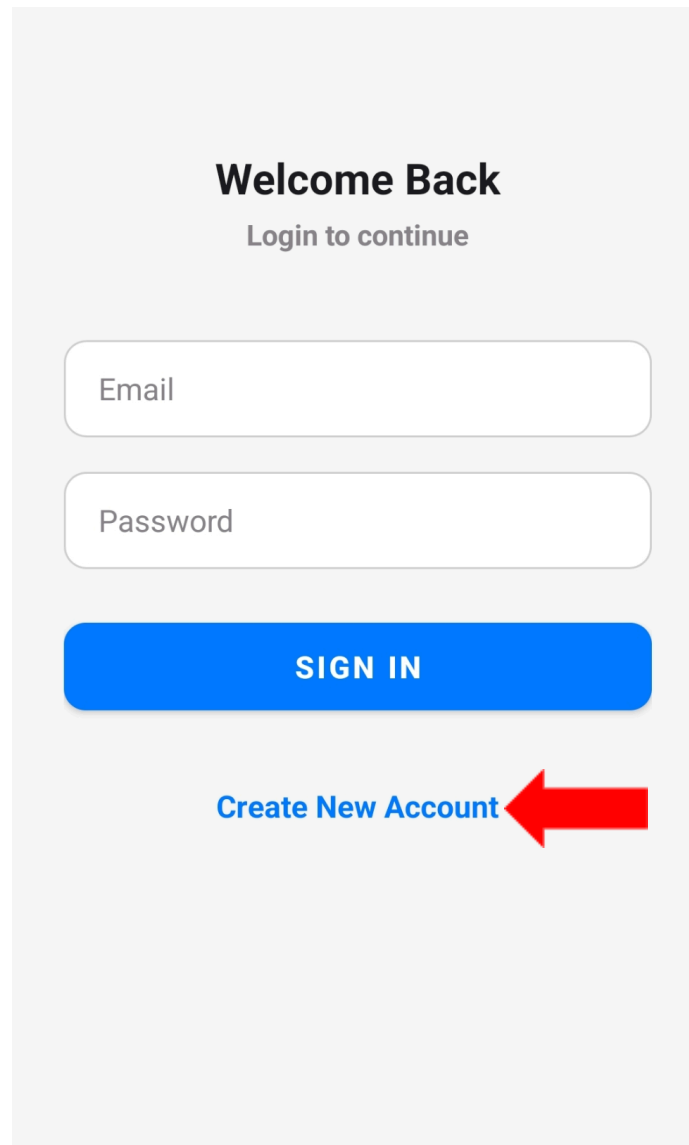


Figure 9 Layout of Chat Screen

## 2.0 Core Workflows

### 2.1 First-Time Setup

Once the SafeChat is downloaded, the user would first be greeted with a **Welcome Back** screen where the user may log into an existing account or create a new one.



The image shows a mobile app screen titled "Welcome Back" with the subtitle "Login to continue". It features two input fields for "Email" and "Password". Below these is a blue "SIGN IN" button. At the bottom, there is a blue link "Create New Account" with a red arrow pointing to it from the right.

**Welcome Back**  
Login to continue

Email

Password

**SIGN IN**

[Create New Account](#)

Figure 10 Welcome Back Screen

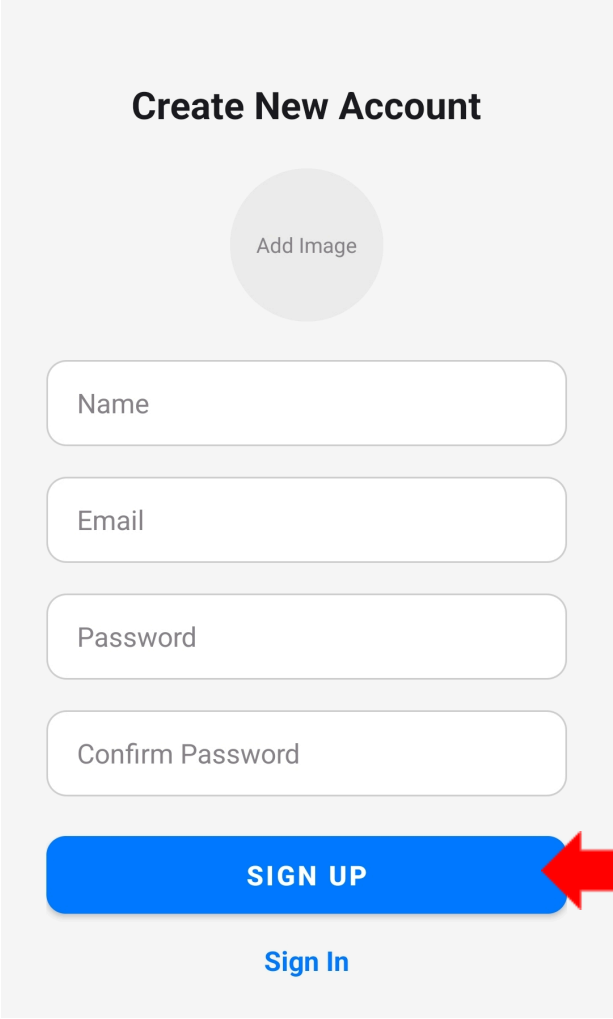
## Step 1: Account Creation and Login

To access the active conversation dashboards and testing environments, users must log in or sign up

### 1. Creating a New Account

If this the user's first time opening the app, user must register an account:

- On the **Welcome Back** screen, tap the blue text link at the bottom labeled **Create New Account** (refer Figure 10).
- This opens the **Create New Account** screen.



The image shows a mobile app screen titled "Create New Account". At the top, there is a circular placeholder for a profile picture with the text "Add Image" inside. Below this are four rounded rectangular input fields labeled "Name", "Email", "Password", and "Confirm Password". At the bottom, there is a prominent blue button with the text "SIGN UP" in white, and a red arrow points to it from the right. Below the button is a blue text link that says "Sign In".

Figure 11 Create New Account Screen

- Tap the circular **Add Image** avatar zone at the top if the user wishes to upload a placeholder profile picture.
- Fill in the required registration text fields:
  - **Name:** Enter testing persona name (e.g., Chong Yu Ming).
  - **Email:** Input target testing email format (e.g., test@gmail.com).
  - **Password:** Create a session password.
  - **Confirm Password:** Re-enter the exact password to verify accuracy.
- Tap SIGN UP to create your account.
- Now users are all set to start chatting!

## 2.2 Primary Functions

### 2.2.1 Signing In to an Existing Account

Once user account has been successfully created, or if users are logging back into a pre-existing profile:

1. Navigate to the **Welcome Back** screen . If users are on the **Create New Account** page, simply click the **Sign In**.
2. Enter registered account email in the first input box.
3. Enter account password in the second input box.
4. Tap the solid blue **SIGN IN** button. User will be logged in and taken to your chat list

**SafeChat** separates itself from standard text clients by constantly monitoring conversation arrays for online grooming signs or manipulation patterns.

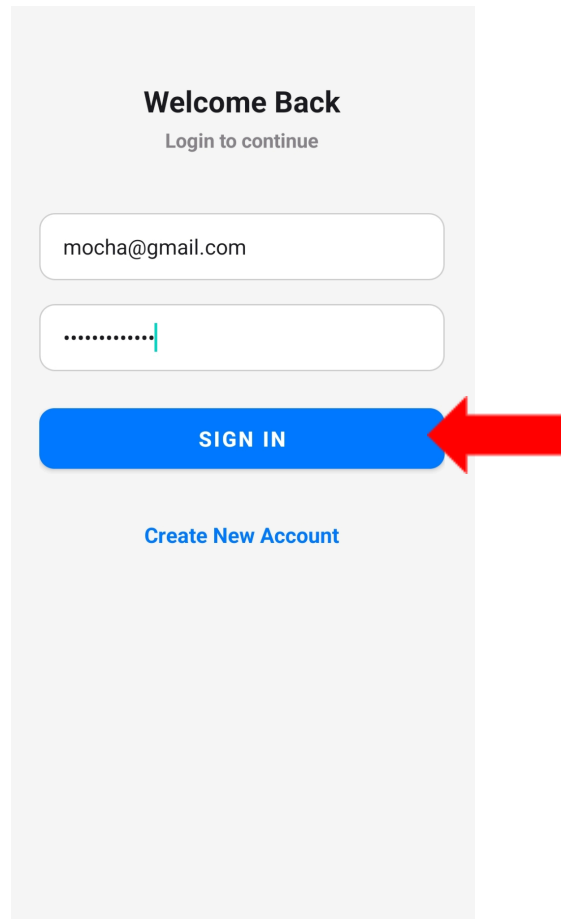


Figure 12 Login Screen

### 2.2.2 Real-Time Risk Flagging

**SafeChat** scans messages for warning signs of grooming. These risks are then categorized as **Medium Risk** or **High Risk**, depending on the level of severity and appropriateness of the text or keywords identified.

#### **Flagging Medium Risk Text**

When incoming messages match known grooming phrases (e.g., secrecy tactics like "*our secret*"), the core application marks the text bubble with a bright red **Warning Triangle Icon** which aims to alert the user.





Figure 13 Chat with Flagged Text  
(Medium Risk)

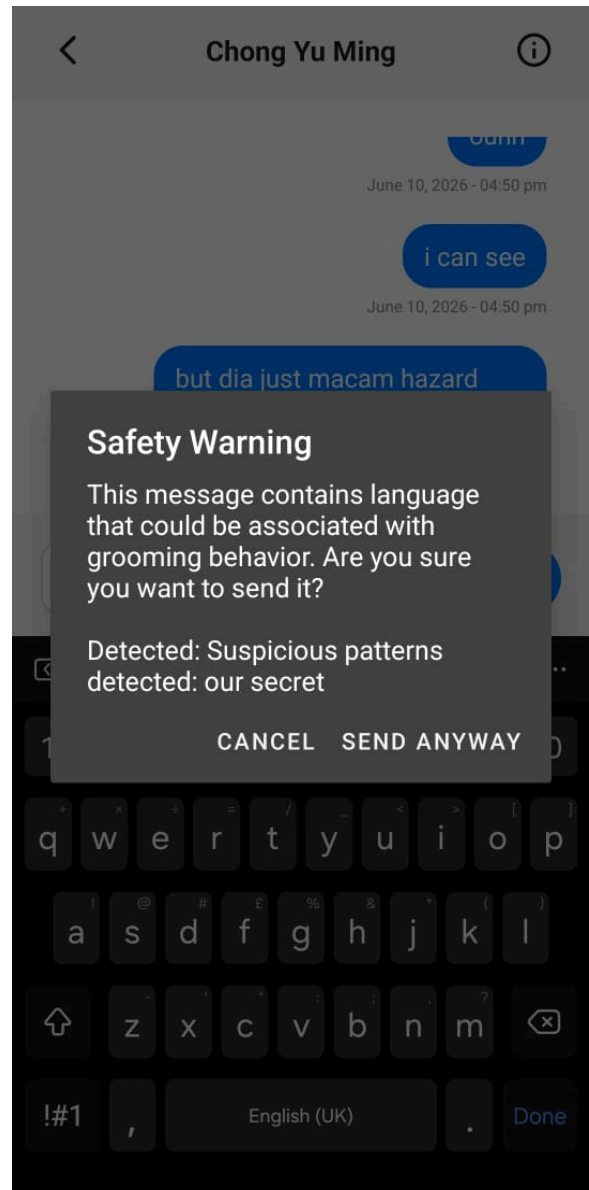


Figure 14 Notification of Safety Warning  
(Medium Risk)

### Flagging High Risk Text

When a message contains phrases or sentences that exhibit suspicious grooming-related linguistic patterns (e.g., elements of isolation such as “Let’s meet alone” ), SafeChat will display a pop-up notification **Security Alert**. The notification blocks the sender and prompts the user to take appropriate actions, such as reporting the incident.

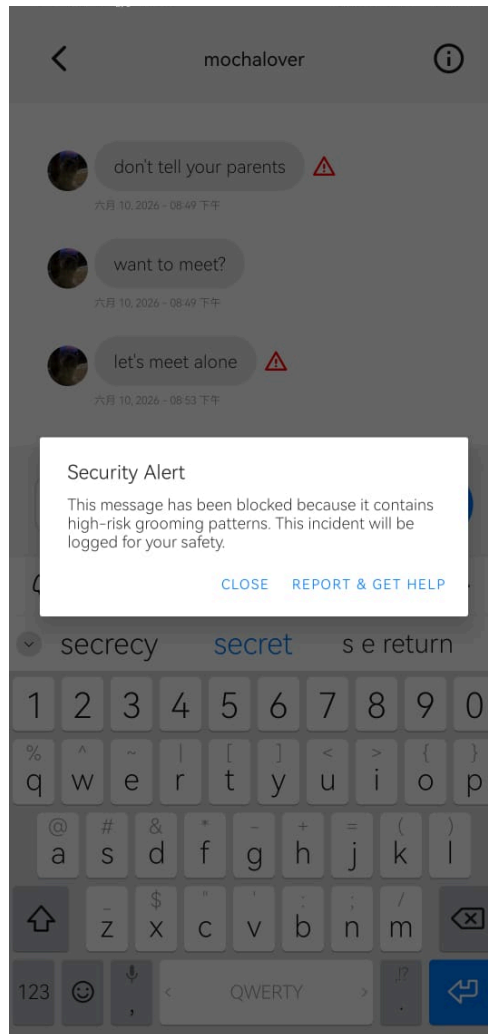


Figure 15 Security Alert (High Risk)

**SafeChat** encourages users to be mindful of the content they send by detecting and flagging potentially inappropriate messages. This feature helps maintain a positive and safe communication environment for all users.

### 2.2.3 Risk Dashboard Assessment

When a conversation begins generating threat indicators, the user or administrator can check how risky the conversation is:

- Tap the **Information ("i") Icon** located in the top-right header area inside the chat with user.
- The system opens the **Risk Dashboard** window.
- This view features a circular graphical progress wheel tracking the active **Risk Score** along with a clear semantic badge classified as **HIGH RISK**.
- Below the visual chart, a dedicated text panel lists the threat rationale: This conversation has multiple high-risk indicators (2 flagged messages). We strongly recommend ending this interaction.
- The app shows you exactly which words triggered the alert: Detected patterns: our secret, don't tell.

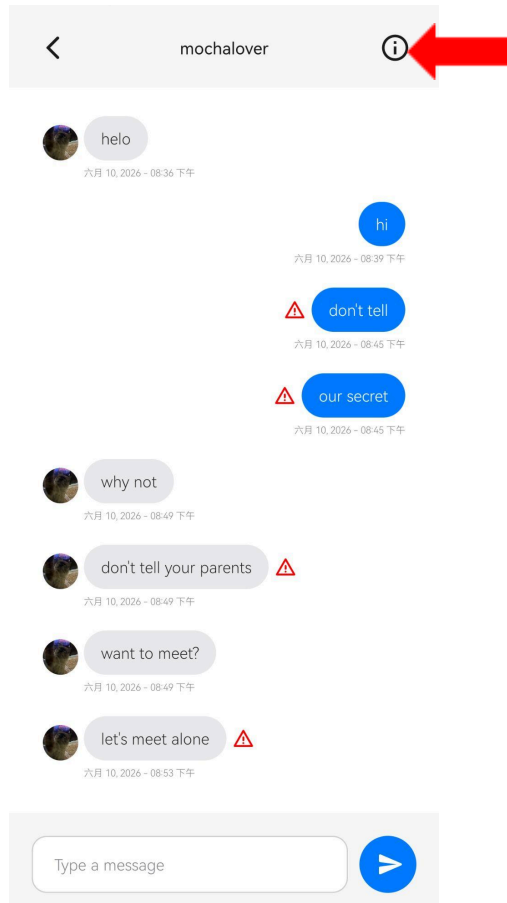


Figure 16 Navigation to Risk Dashboard

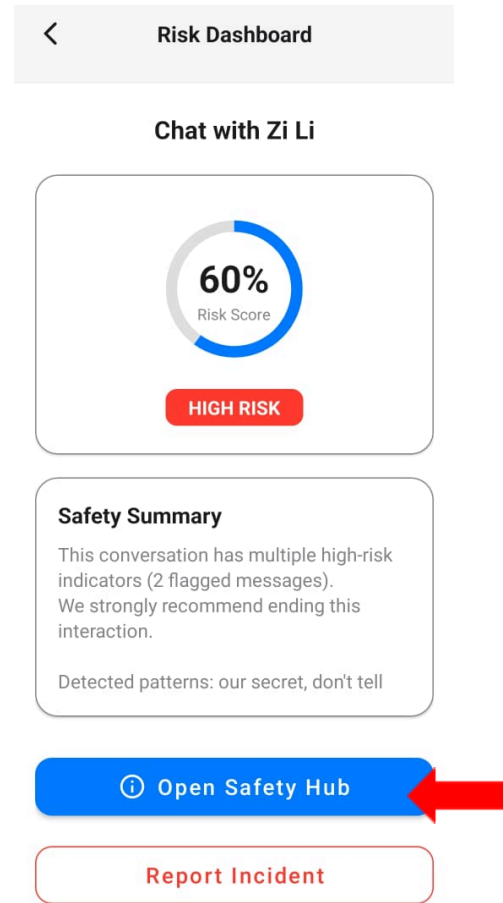


Figure 17 Risk Dashboard

## 2.2.4 Interacting with the AI Safety Hub

From the Risk Dashboard, tapping the blue **Open Safety Hub** button opens the Safety Hub chat assistant:

- The system loads an automated assistant menu stating: How can I help you today?
- Users can query or toggle specific modular workflows using the quick-access footer controls: **Policy**, **Report**, and **Resources**.
- Supported functions include an **Awareness Chatbot**, a **Policy Analyzer** for inspecting suspect terms, a **Report Writing Assistant**, and a **Resource Recommender**.

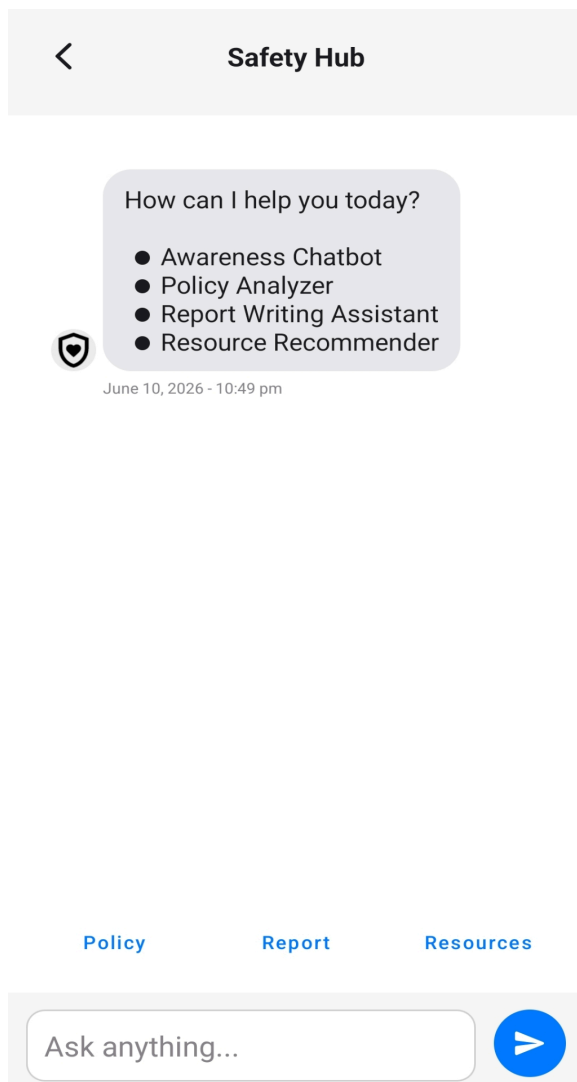


Figure 18 Safety Hub

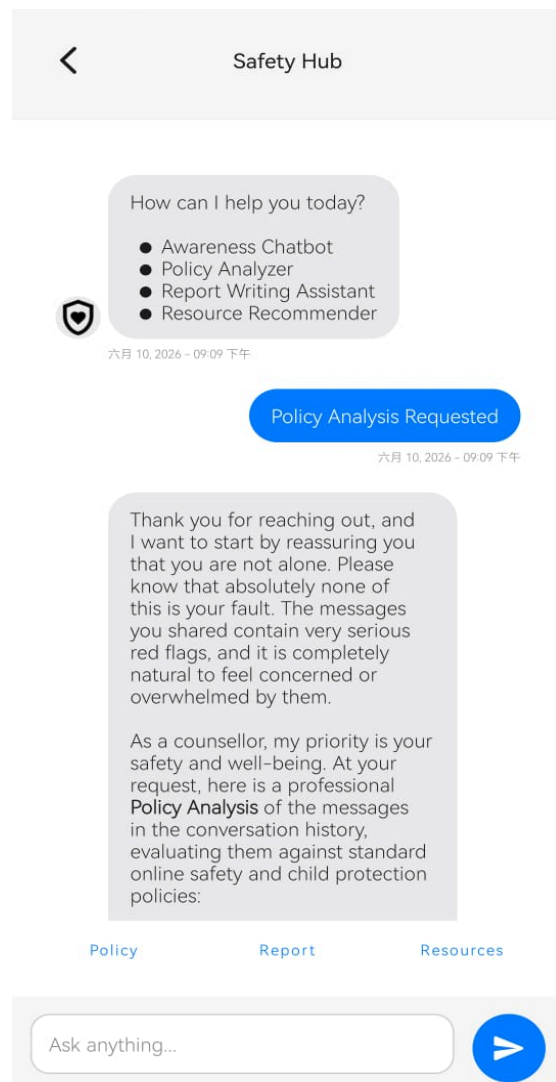


Figure 19 Policy



Figure 20 Report Assistant

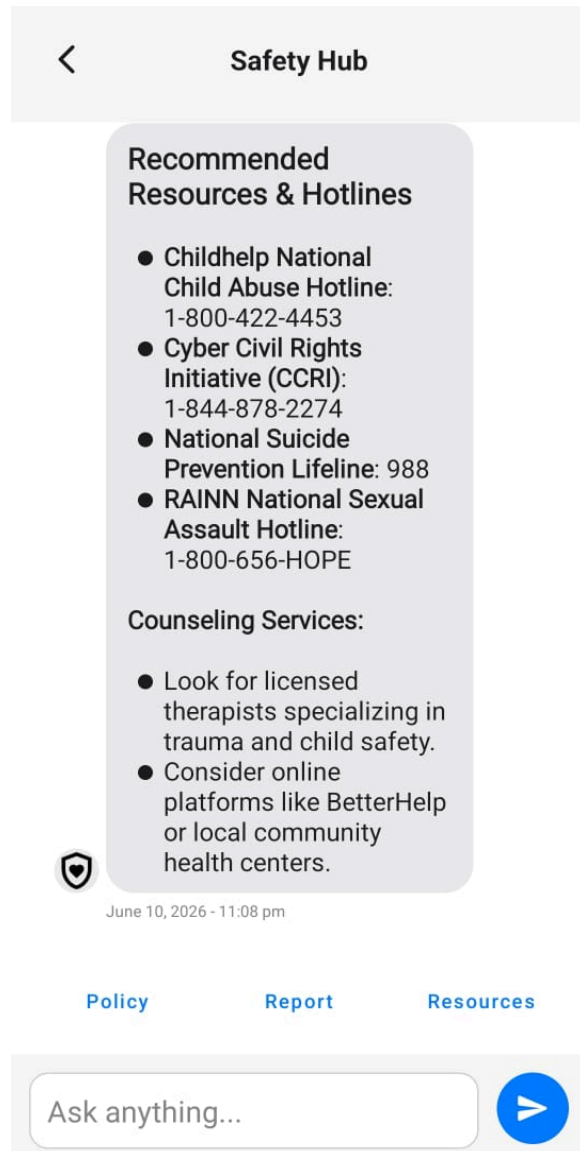
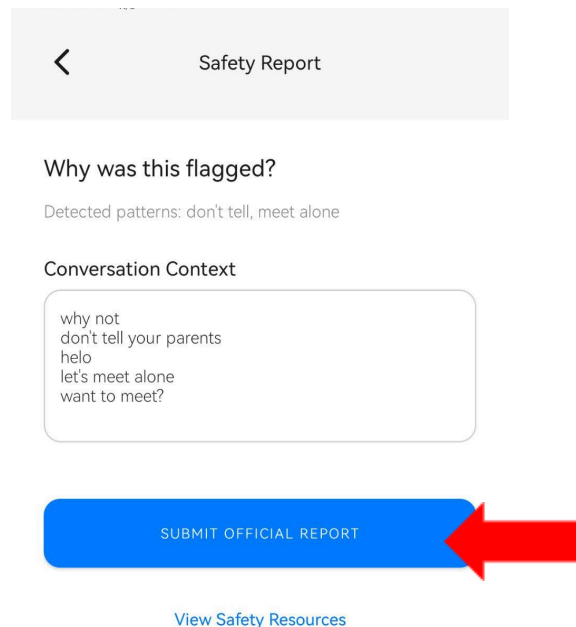


Figure 21 Help Resources

## 2.2.5 Documenting and Submitting Official Incident Reports

If an interaction escalates beyond a safe threshold, users can draft formal documentation directly through the platform:

- Select the **Report Incident** or **Report** action terminal options within the safety panels.
- The application loads the **Safety Report** screen, documented.
- The header details the direct trigger root cause: Why was this flagged? User flagged from dashboard. The user should also be able to input their own details for the message.
- A text container labeled **Your Message** automatically captures the context profile.
- Clicking the prominent blue **SUBMIT OFFICIAL REPORT** bar sends your report for review. Users can alternatively select **View Safety Resources** to exit back to educational safety guidelines.



< Safety Report

Why was this flagged?

Detected patterns: don't tell, meet alone

Conversation Context

why not  
don't tell your parents  
helo  
let's meet alone  
want to meet?

SUBMIT OFFICIAL REPORT

[View Safety Resources](#)

Figure 22 Safety Report

## 2.3 Limitations and Important Disclaimers

Before relying on SafeChat, it is important to understand exactly what the application can and cannot do. SafeChat is a safety **aid**, not a guarantee of safety.

### What SafeChat Is?

SafeChat is a chat application that scans outgoing messages for language patterns commonly associated with grooming behaviour, and provides an AI-powered Safety Hub for education and guidance. It is designed to raise awareness and give users an early warning when a conversation shows concerning signs.

### What SafeChat Is NOT

- **SafeChat is not a certified or clinically validated grooming detector.** It uses keyword matching, not human judgment. It can be wrong in both directions: it can miss real grooming, and it can flag harmless messages.
- **SafeChat is not a replacement for reporting to real authorities.** Submitting a report inside the app does not notify the police or any official body. If you are in danger, you must contact the police, a trusted adult, or your institution's official reporting channels directly.
- **The Safety Hub chatbot is not a licensed counsellor.** It is an AI assistant. Its responses may be incomplete, generic, or occasionally inaccurate. It should not be treated as professional psychological, legal, or medical advice.
- **SafeChat is not a monitoring tool for parents or guardians.** It only flags messages to the person sending or receiving them; it does not notify a third party automatically.

### Known Limitations

- **It can miss real grooming behaviour.** SafeChat only checks for specific words and phrases. A person trying to groom someone can avoid detection by using misspellings, coded language, slang, or by spacing risky words across multiple messages.
- **It can flag harmless messages.** Because detection is based on matching words rather than understanding context, normal conversations can sometimes be flagged. For example, "don't tell anyone about the surprise party" may trigger a warning even though it is harmless.
- **It currently only checks messages you send, not messages you receive.** If someone else is sending you concerning messages, those messages are not automatically scanned in real time. (The Risk Dashboard does look back at the other person's messages, but this happens only when you choose to open it.)
- **It only works in English.** Messages in other languages, or mixed-language slang, are not reliably detected.

- **The Safety Hub sends your conversation text to Google's Gemini API** to generate its responses. While this is required for the feature to work, it means message content leaves your device for processing.

### **If You Are in Immediate Danger**

If you ever feel unsafe, threatened, or are being pressured, manipulated, or asked to keep secrets by someone you are chatting with:

1. **Stop the conversation** and do not meet the person in real life.
2. **Tell a trusted adult, friend, or authority figure immediately** — do not rely on SafeChat alone.
3. **Contact your local police or emergency services** if you believe you or someone else is in danger.
4. You may also use the **Report Incident** feature in SafeChat to document the conversation, but treat this as a supplement to reporting elsewhere, not a substitute for it.

SafeChat is here to help you recognise warning signs and feel less alone — but the responsibility for your safety should never rest on an app alone.



## 3.0 Settings and Customization

### 3.1 Profile Management

The configuration settings in this prototype let users customize their profile settings:

- **Updating Account Metadata:** From the main workspace configuration menu, users can adjust their display name (e.g., changing how their identity displays in the active hub header bar).
- **Managing Session Avatars:** Users can modify or update their account image placeholder by tapping the profile picture management circle, which updates how they appear to other users across the **Select User** directory screen.

### 3.2 App Preferences

To accommodate varying testing environments the application's local performance configurations can be customized:

- **Notification Toggles:** Enable or disable real-time push alerts for incoming messages and urgent risk-flag triggers.
- **Local Database Maintenance:** Because this is a prototype that logs message arrays to calculate risk metrics locally, users can clear chat history to free up space or start fresh.

## 4.0 Support and Troubleshooting

### 4.1 Frequently Asked Questions (FAQ)

1. Does the chat forward to the Police department?

*No. SafeChat does not automatically share your data with the police or any other authority, and your messages are never visible to other users besides the person you're chatting with. However, when you use AI features like the Safety Hub or Report Assistant, your conversation text is sent to Google's Gemini API for processing. This happens automatically whenever you use these features, with no separate opt-in step. This is necessary for those features to work, and is handled in line with the Personal Data Protection Act 2010 (PDPA).*

2. Does the application require an active internet connection?

*SafeChat relies on remote servers to handle user registration, user directory updates, and real-time text delivery. A continuous data or Wi-Fi connection is mandatory to keep the database synced.*

3. Do I have to grant the application permission to send notifications?

*No, SafeChat can function without granting notification permission. However, enabling notification permissions is highly recommended to continue receiving chat notifications when not in use.*

4. What should I do if the application layout does not resize properly when typing?

*If SafeChat misaligns, simply press the back button to hide the keyboard and tap the text field again to reset the layout view.*

5. What does the "Safety Warning" alert mean when I try to send a message?

*SafeChat features an automated scanner that detects dangerous text. If your message contains identified grooming behaviors, the alert triggers to allow reviewing flagged text before sending.*

6. Can I still send a message if it triggers a standard Safety Warning?

*Yes, SafeChat allows you to make the final choice for standard warning. You can still bypass the warning and deliver the message, or tap "Cancel" to return to the chat screen and rephrase your text.*

7. Why does a "Security Alert" completely block certain messages from being sent?

*When the internal filter detects a text containing severe and high-risk grooming patterns, it*

*activates a hard block. The message is terminated before it reaches the server to protect the safety of the user and the incident is logged locally. You can also choose to REPORT AND GET HELP (View FAQ Question 9)*

8. How can I seek help or learn more about digital safety within the application?

*You can tap "Open Safety Hub" to access a dedicated security panel. It is located at the top right of the main page, shown as a wrapped "i" icon. It features Automated Awareness Chatbot that allows users to ask questions to chatbot and receive answers, Policy Analyzer, Report Writing and also Resource Recommender.*

9. How do I report an abusive user or an incident to platform administrators?

*Navigate to the reporting section of the application, input the details of the event under the "Your Message" field, and tap "Submit Official Report" to route the file to management.*

10. Where can I view my profile metrics and account security status?

*You can open the Risk Dashboard to review your personalized metrics (Located inside the chat between users). It displays your live Risk Score with an automated Safety Summary based on chat interactions. You can also open the Safety Hub or report incidents within the chat.*

11. What should I do if the application encounters a network sync error or freezes?

*Close the application completely, go to Settings > Apps > SafeChat and tap 'Clear Cache' before relaunching the software.*

## 4.2 Error Messages

When testing the **SafeChat** application bundle on customized device environments (such as Android Virtual Devices or Xiaomi mobile ecosystems), you may encounter the following system alerts:

| Error Message / System Behavior | Root Cause | Corrective Action |
|---------------------------------|------------|-------------------|
|                                 |            |                   |

|                                                              |                                                                                                                                                            |                                                                                                                                                                                                                                          |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>"Blocked by Play Protect" or "Unrecognized Developer"</b> | The Android operating system flag triggered because the prototype is manually sideloaded as an APK from GitHub rather than a commercial app store.         | Tap the <b>More Details</b> drop-down arrow inside the system warning pop-up, then select <b>Install Anyway</b> to bypass the block.                                                                                                     |
| <b>"Installation Forbidden" (Android Devices)</b>            | Android system's native security layer blocks manual application package execution from third-party app origins by default.                                | Navigate to <b>Settings &gt; Privacy Protection &gt; Special Permissions &gt; Install Unknown Apps</b> . Grant explicit permission to your web browser or file manager, acknowledge the 10-second warning countdown, and tap <b>OK</b> . |
| <b>Risk Score / Flag Missing</b>                             | The local device network is experiencing lag or is disconnected from the processing node, preventing the pattern detection engine from parsing text rules. | Verify your internet connectivity. Ensure your environment matches the recommended specifications (zero emulated network latency) to allow real-time keyword parsing.                                                                    |
| <b>"Invalid Credentials" on Welcome Back Screen</b>          | The email or password entered does not match an authenticated test persona stored within the application's active database.                                | Check your spelling in the input text blocks. If you have not created an identity yet, tap <b>Create New Account</b> at the bottom of the screen to register your test profile first.                                                    |

Table 1 Error Messages

## 4.3 Contact Info

For further information about the use of **SafeChat**, users may reach out to us personally through one of the methods below:

### 4.3.1 Project Manager and Overseer

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### 4.3.2 User Manual Developers

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### 4.3.3 Other Contributors (Programming/Testing/Design/Developer Manual etc.)

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